First, you need to add a dependency on the Testcontainers module for PostgreSQL in the build.gradle file of your Catalog Service project. Remember to refresh or reimport the Gradle dependencies after the new addition.

Listing 5.16 Adding dependency on Testcontainers in Catalog Service

```
ext {
  ...
  set('testcontainersVersion', "1.17.3")                    ❶
}

dependencies {
  ...
  testImplementation 'org.testcontainers:postgresql'        ❷
}

dependencyManagement {
  imports {
    ...
    mavenBom "org.testcontainers:
    ➥ testcontainers-bom:${testcontainersVersion}"         ❸
  }
}
```

❶ Defines the Testcontainers version to use

❷ Provides container management features for PostgreSQL databases

❸ BOM (bill of materials) for Testcontainers dependency management

When running tests, we want the application to use a PostgreSQL instance provided by Testcontainers rather than the one we configured earlier via the `spring.datasource .url` property. We can overwrite that value in a new application-integration.yml file created under src/test/resources. Any property defined in this file will take precedence over the main one when the `integration` profile is enabled. In this case, we'll overwrite the value for `spring.datasource.url` following a format defined by Testcontainers.